

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. MTB-609: Relativity II

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (@) A proton of rest mass 1.67×10^{-27} grams is moving with velocity $0.995c$. Calculate its energy and momentum. [4]

- (a) Show that the momentum of a particle of rest mass m_0 and kinetic energy K is given by $[K = \gamma m_0 c^2 - m_0 c^2]$ [4] D/4(303)—600 (Turn Over)
- (b) Obtain the transformation equations for H . [4] ies ain en [2]
- (c) Prove that $E^2 - H^2$ is Lorentz invariant. [4] . 5 => 3
- (d) Find F_x and F_y , in terms of E and H . [4]
- (e) State the expression for the Lorentz force density. [2]

Question 2. (a) Establish the transformation formula for mass and hence prove that $\gamma > 1$ [6]

- (a) [6]
- (b) Two particles with proper masses m_1, m_2 move along the x-axis of an inertial frame with velocity u_1, u_2 respectively. They collide and coalesce to form a single particle. Assuming the laws of conservation of relativistic momentum and energy, prove that the proper mass M , and velocity u of the resulting single particle are given by : $M^2 = m_1^2 + m_2^2 + 2m_1 m_2 \gamma(u_1 - u_2) = M_1^2 + M_2^2 + 2M_1 M_2 \gamma(u_1 - u_2)$ [6]
- (c) ce? [6] D/4(803) (Continued)

Question 3. (a) Establish the transformation formula for energy and momentum. [6]

- (a) What do you understand by relativistic Hamiltonian? [6]

Question 4. (a) Obtain the transformation formulae for the densities of electric charge and electric current. [6]

- (a) Prove the Lorentz invariance of Maxwell's equations without using tensors. [6]

Question 5. (a) Obtain the transformation equations for A_μ and ∂_μ , where A_μ is electromagnetic vector potential. [6]

Find 'all the components of anti-symmetric tensor $F_{\mu\nu}$ in four dimension, in terms of E and H . [6]

Question 6. Obtain the expression for the electromagnetic energy-momentum tensor in Minkowski space, 12

Question 7. (a) Write a short note on the equivalence of mass and energy. [6]

(a) Using Lorentz condition, show that $\partial_\mu A^\mu = 0$ and $\partial_\mu \partial^\mu A^\nu = 0$

(b) 6 . wake D/4(303)—600 BS/Sem VI/208